

Scalability & Future Plan

Date	03-07-2026
Team ID	SWTID-2026-3845
Project Name	SMART LENDER
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No batch processing; applicants must be evaluated one at a time	Slows down high-volume evaluation for financial analysts	High
2	Model not yet deployed on IBM Cloud; runs only locally	Limits accessibility for bank staff outside the local environment	High
3	No automated document verification for flagged high-risk applicants	Requires manual follow-up, reducing efficiency gains	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user local Flask instance	Deploy on Render Cloud with load balancing to support multiple concurrent users
2	Data Storage	Applicant data processed in-memory, not persisted	Integrate a cloud database (e.g., IBM Db2 or Cloudant) to store applicant records and predictions
3	Performance	Model loaded and predictions served synchronously	Implement caching and asynchronous request handling for faster response under load
4	Security	No authentication or encryption implemented	Add user authentication, HTTPS, and encryption for sensitive applicant data (income, credit history)

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Deploy application on Render Cloud with a live public URL	1–2 months post-launch	Enables real-time access for bank staff across locations, improving adoption
Phase 3	Add batch evaluation feature for uploading and processing multiple applicants at once	2–3 months post-launch	Significantly reduces evaluation time during high-volume periods for financial analysts
Phase 4	Integrate automated high-risk flagging with document verification workflow	3–5 months post-launch	Strengthens fraud/risk detection and secures sensitive applicant data, increasing institutional trust